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To cite this article: Septafiansyah Dwi Putra *et al* 2022 *IOP Conf. Ser.: Earth Environ. Sci.* **1012** 012090

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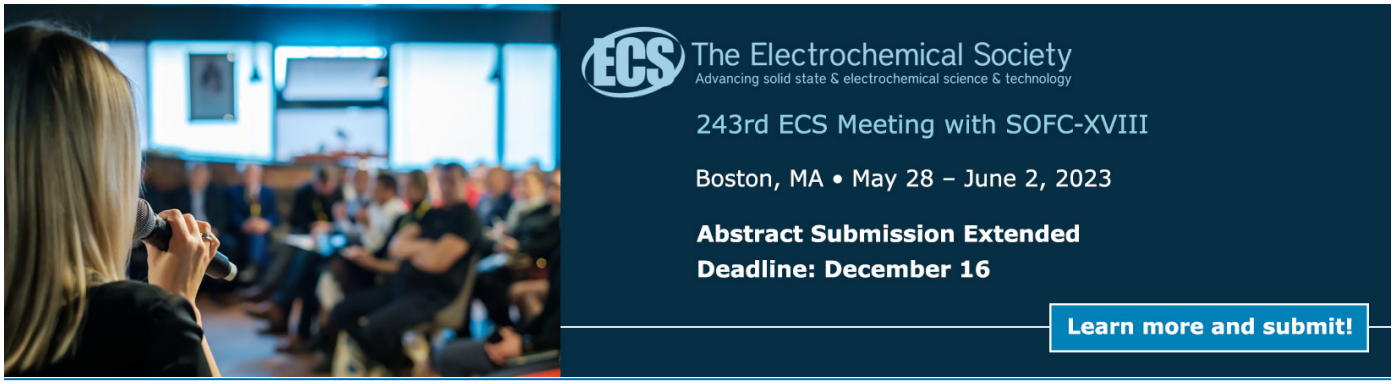

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Design of IoT Monitoring System Based on LoRaWAN Architecture for Smart Green House

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Abstract. The application of smart agriculture as an applied innovation in overcoming uncertainty over natural conditions and supporting national food security. The utilization of smart agriculture with the Internet of Things (IoT) approach can spur increased added value, competitiveness, and commercial benefits for national food products. In this paper, we proposed a low-cost, modular, and Long-Range Wide-Area Network (LoRaWAN) monitor and collect the environmental data (air temperature and humidity) related to the growth in the smart greenhouse over a while. The proposed architecture has been evaluated on a real farm on a lab scale with RSSI -118 and SNR for 2% for 1000m. A web-based visualization tool for the collected data is also presented to validate the LoRaWAN architecture.

1. Introducing

The agricultural top issues include population growth, massive resource exploitation, climate change's impact on agricultural productivity, and the threat of natural disasters [1]. As a result, an approach based on applied innovation is required to overcome uncertainty about natural conditions. There is a new approach to addressing the impacts of climate change and other natural disasters, namely the concept of smart agriculture. The concept of smart agriculture emphasizes the importance of farming by monitoring the condition of agricultural land, automating processes, digitizing connectivity, and making precise decisions based on accurate data and information. Smart agriculture with an Internet of Things (IoT) approach is required in the food sector to ensure the highest quality agricultural food products. The utilization of ICT, integrated with sensors and automation in real-time, will provide accurate data and information to be evaluated together in making decisions. The decision's outcome is immediately applicable to farmers and other stakeholders. Thus, the three primary objectives of smart agriculture are to increase the value-added, competitiveness, and commercial benefits (profit/benefit) of domestic food products and contribute to national food security.

This research was designed to produce a system that was able to monitor and collect environmental conditions with various IoT sensors, digitizing environmental information with dashboard systems and mobile devices. The research applies various technological innovation approaches such as sensor technology, LoRa (Long Range) network communication systems in agricultural environments, environmental monitoring systems, data collection, data processing, and visualization techniques to help provide the best decisions in a smart greenhouse environment. So that problems in previous studies such as digitizing environmental information with a dashboard system can be resolved. Greenhouse technology is the application of agricultural technology. This technology provides a controlled environmental situation for plants. To protect plants from various extreme conditions such as weather changes, cold, extreme temperatures and various pests that may arise. The concept of

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sustainable agriculture through greenhouse technology has been developed in various studies. A study revealed by Azaza in 2016 stated that the use of smart greenhouse technology could reduce energy and water savings by 22% - 33%, so that the use of smart greenhouses supports a sustainable agricultural environment [2,3]. Intelligence monitoring system also already state at [4] by using ZigBee and CC2530 chips as controller. Meanwhile, the LoRa protocol is a new approach that enables low-power [5], long-distance and high-performance data transmission [6], and can be implemented in smart greenhouse technology.

2. Related Work

2.1. IoT and the key to supporting smart agriculture

Internet of Things (IoT) represents the use of internet connectivity to support everyday life. The term IoT which Kevin Aston introduced, continues to increase. IoT has become a global term as defined by ITU-T, Auto-ID laboratories, and IERC. However, conceptually it is described as "a worldwide network of networked devices that are uniquely addressable via established communication protocols.". The use of IoT in monitoring and automation in the agricultural sector will continue to increase along with the large potential for research and application of the technology. Especially in 2050, it is projected that the agricultural sector with IoT will increase food production by 70% and serve 9.6 billion people, and 2 billion sensors are applied to 525 million agricultural lands [7].

Meanwhile, in a deeper aspect, smart agriculture is defined as a form of diversification of condition monitoring, integration, storage, analysis, remote control and monitoring, and processing and sharing of information in decision making on the management of the agricultural sector using technology [4]. In contrast to conventional agriculture (traditional agriculture) is manual management both before harvesting, development, and post-harvest, which requires a large amount of time, large labour, and high levels of uncertainty over agricultural products [4]. The real implementation of IoT and embedded systems in the agricultural sector is shown in various systems, namely soil and water monitoring [8], land monitoring [9], irrigation management [10], disease and pest monitoring [11], seed control [12], migration irrigation systems [13], intelligent greenhouses [14], remote diagnosis and control for agricultural gardens [8], and asset control of agricultural equipment [8].

2.2. LoRA and LoRaWAN

LoRa is an abbreviation for Long Range. Semtech developed this wireless communication technology. This communication technology is primarily used in machine-to-machine (M2M) and Internet of Things (IoT) networks. Long-range capability, low power consumption, and high robustness are just a few of the key characteristics of LoRa. A single gateway or base station can cover an entire city or hundreds of square kilometres. LoRa's physical layer is based on frequency shift keying (FSK) modulation. Modulation is a very efficient method of achieving low power consumption. LoRa employs a spread spectrum technique known as Chirp Spread Spectrum (CSS), which retains the low-power characteristics of FSK modulation while significantly increasing communication range. CSS has been used in military and space communication for decades due to its ability to achieve long communication distances and resistance to interference. LoRa is the first low-cost commercial implementation. The simplest explanation for CSS is that each bit of information is encoded in the transmitted signal as multiple chips. LoRa utilizes several orthogonal spreading factors (SF). SF values range from 6 to 12, but Semtech does not currently support using the spreading factor 6. SF allows for a trade-off of data rate and range. As illustrated in figure 1, increasing the spreading factor increases the range but decreases the coverage probability and vice versa.

3. Methods

The purpose of LoRAWAN for smart agriculture is connecting the sensor and LoRA gateway in the rural farm feld to the control center for accuttion data. The method was used in this research are developmental process that includes the researcher's assumptions, values, and conditions for analyzing the data. The generic method used in this paper is proposed by [16]. The reason for choosing this

method is that it allows for iteration, which means that previous steps of the method can be updated and edited, as shown in figure 1. The researcher is given the opportunity to make modifications and changes, thereby acquiring new knowledge.

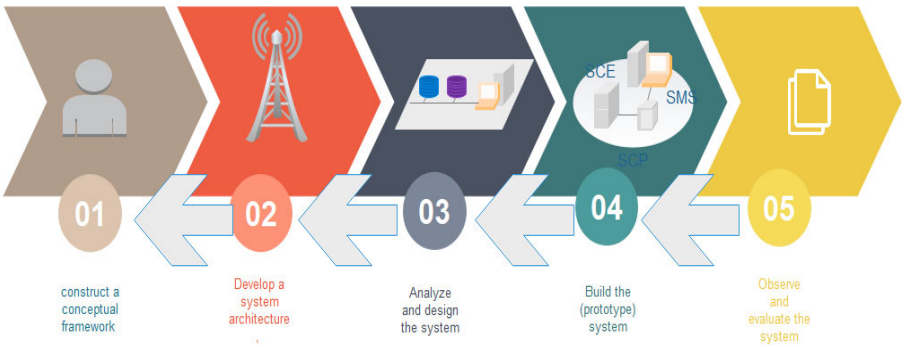

Figure 1. Research Method

The proposed conceptual framework illustrated the models' LoRa and sensor configuration conditions. With a point to point setup, the LoRa network establishes a network with a shorter distance. The model demonstrates two different types of LoRa sub nodes and two different types of integrated LoRa with sensors communicating with a single LoRa gateway. The device is used to measure distances of 100m, 300m, 600m, and 1000m in the experiment. When data from all LoRa sub nodes was transmitted 100 times for each ID sub node, there was a data collision. The same thing occurred with the experiment combining LoRa and DHT sensors. The dashboard monitoring system will display all received data.

Develop a system architecture. Four primary components may be identified that must interact effectively in order to obtain the intended results: (1) The LoRa sender equipped with sensors, which will be used to assess the greenhouse's environmental conditions. (2) The LoRa gateway and infrastructure, which functions both as a gateway, receiving data from other gateways, and as a MQTT sender, capable of sending collected data to other MQTT-compliant services. (3) The dashboard system, which will accept data from the LoRa gateway and store it for study in addition to the existing services. (4) The device used by the end user or administrator to evaluate the entire system. The figure 3 depicts the elements' generic interrelationship. The system architecture for this research as shown in figure 2

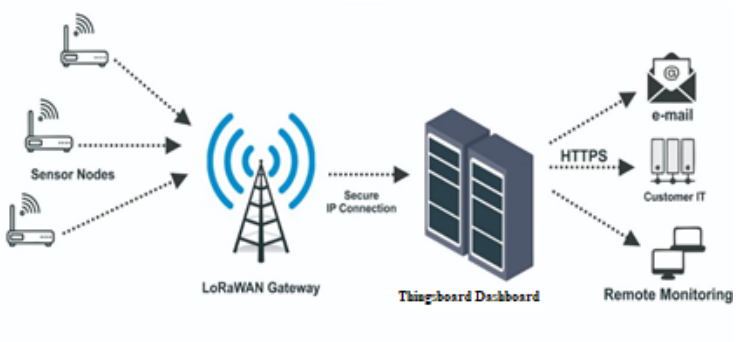

Figure 2. System architecture designed

Analyze and design the system. Since there is a high correlation between weather, water, and soil parameters variation in the smart greenhouse, and because changing one of these parameters affects the other two, combining them all into a single monitoring module is a nice idea. This module is

comprised of the following components: LoRa sender, LG02 dragino, DHT22 sensor, soil moisture sensor, and MQTT server running on Linux server. The design of the system is shown in figure 3.

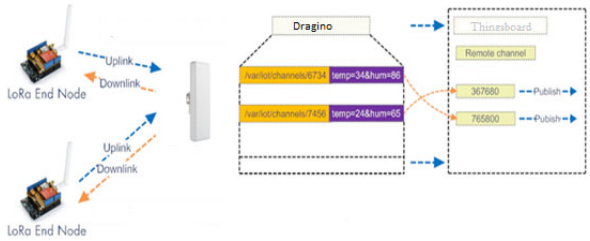

Figure 3. LoRa and MQTT server designed

Build the (prototype) system. The circuit made is placed in a box built on ESP32, The LoRa SX1276 and DHT22 devices and LoRa antennas. The design of the black box that will be used for each node can be seen in Figure 4. The dashboard system is based on the output of the DHT22 sensor; this sensor will monitor the climate's temperature and humidity and send an alert or notification to the system if one of these values exceeds the threshold.

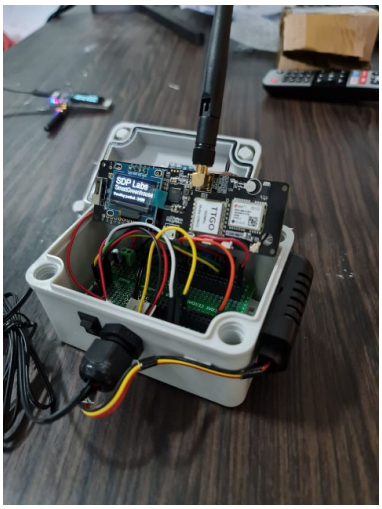

Figure 4. The prototype of LoRa and sensor DHT22

Observe and evaluate the system. To evaluate the LoRa architectural system in this smart greenhouse, two LoRa node devices are connected point to point and LOS (line of sight). Changes to the distance parameter will affect several values such as RSSI (Received Signal Strength Indicator), SNR (Signal Noise Ratio), and PDR (Packet Delivery Ratio), which show the results of the quality of the LoRa transmission media, as well as PLE. Some of the functions to finding the value of some of the evaluation parameters above are:

RSSI (Received Signal Strength Indicator) is an evaluation indicator of signal strength received by LoRa receivers. RSSI is considered good from how far the closed RSSI is from the value 0. If it is close to the value 0, then the RSSI can be said to be good. If it is getting further away from 0, then the RSSI is said to be bad. A good RSSI value ranges from "-1" to "-99" The average RSSI can be determined through:

$$RSSI = \frac{Number\ of\ RSSI}{Received\ packet}$$

SNR (Signal Noise Ratio) is a comparison between signal strength and noise level. The greater the SNR value, the better the quality of the channel. The SNR value is obtained by calculating:

$$SNR = \frac{Number\ of\ SNR}{Received\ packet}$$

4. Finding and Discussion

Each LoRa and sensor node is based on a TTGO LoRa V2.1 915Mhz module and an ESP32 with an SX1276 and GPS module. This board supports LoRa and is well-suited for low-power applications. It is a cost-effective and practical solution for LoRa-based IoT applications. Temperature and humidity sensors DHT22 are used exclusively for sensing and data generation. They are connected to TTGO boards and the dragino OLG2 LoRa gateway directly. These sensors are setup via the 915 MHz antenna after being connected to the LoRa gateway. This antenna is utilized for data transmission between the sensor and the gateway.

Numerous analytical activities were conducted in conjunction with this research. This includes an examination of spread ranges, power consumption, the performance of software analysis, and network performance. The RSSI and SNR were used to analyze network performance in this article. Numerous data collecting things need to be considered to ensure that LoRa sub-nodes may transmit data to the LoRa gateway at any time without experiencing severe packet loss. This enables acceptable wireless network operation with a low likelihood of frame collision. It is critical to test the sub nodes' ability to transfer large amounts of data without clashing. The measurement of RSSI and SNR are shown in Table 1 and Figure 5.

Table 1. Measurement of RSSI and SNR

No	Distance Length	Min RSSI	Max RSSI	Min SNR	Max SNR
1	100 m	-104	-90	8,75	11,25
2	200 m	-111	-103	8	11,25
3	300 m	-115	-102	7	9,25
4	600 m	-122	-113	-3,25	7
5	750 m	-124	-116	-7	4,25
6	1000 m	-125	-118	-7,75	2

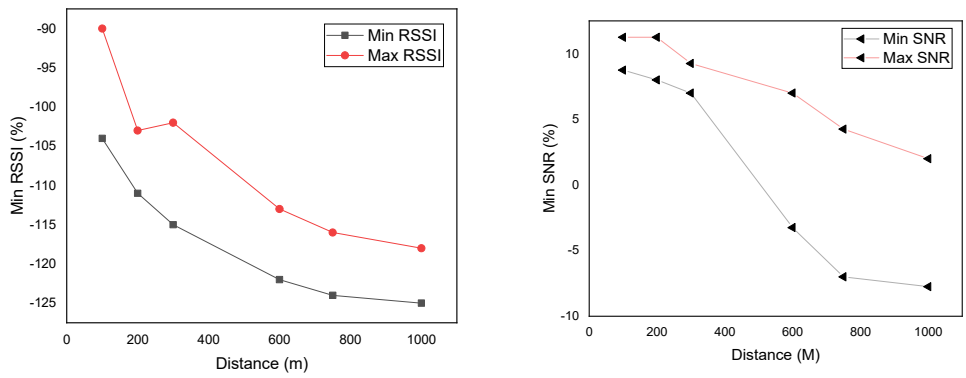

Figure 5. The RSSI and SNR Value Measurement

The RSSI (Received Signal Strength Indicator) and SNR values in Figure 5 represent the strength of the received signal and the gateway's ability to receive it from the IoT node. The low RSSI values observed are due to packets being broadcast by IoT nodes located near the gateway.

5. Conclusion

We integrated LoRaWAN sensor support within the smart greenhouse in this paper by leveraging existing hardware and services. More precisely, we designed a LoRaWAN protocol middleware architecture that adheres to numerous best practices. The middleware was designed with extensibility in mind, making it simple to work with various data sources and formats. Additionally, we worked within the thingsboard dashboard services to ensure that the middleware service ran properly. Future research will focus on extending complete automation by including other energy optimization strategies tailored to LoRaWAN, such as deep sleep mechanisms, security[17], and cognitive capabilities[18].

Acknowledgement

The work was supported by the Ministry of Education, Culture, Research and Technology of the Indonesia in the framework of contract agreement 255/E4.1/AK.04.PT/2021 – the derivative contract number is 380.2/ PL15.8/PT/2021.

References

[1] Azaza M, Tanougast C, Fabrizio E and Mami A 2016 Smart greenhouse fuzzy logic based control system enhanced with wireless data monitoring *ISA Trans.* 61 297–307

[2] Achour Y, Ouammi A and Zejli D 2021 Technological progresses in modern sustainable greenhouses cultivation as the path towards precision agriculture *Renew. Sustain. Energy Rev.* 147 111251

[3] Wu Y, Li L, Li M, Zhang M, Sun H, Sygrimis N and Lai W 2019 Remote-Control System for Greenhouse Based on Open Source Hardware *IFAC-Pap.* 52 178–83

[4] Chen J and Yang A 2019 Intelligent Agriculture and Its Key Technologies Based on Internet of Things Architecture *IEEE Access* 7 77134–41

[5] Nakamura K, Manzoni P, Redondi A, Longo E, Zennaro M, Cano J-C and Calafate C T 2021 A LoRa-based protocol for connecting IoT edge computing nodes to provide small-data-based services *Digit. Commun. Netw.*

[6] Ram Prasanna K, Mathana J M, Ramya T A and Nirmala R 2021 LoRa network based high performance forest fire detection system *Mater. Today Proc.*

[7] Ahmed N, De D and Hussain I 2018 Internet of Things (IoT) for Smart Precision Agriculture and Farming in Rural Areas *IEEE Internet Things J.* 5 4890–9

[8] Ojha T, Misra S and Raghuwanshi N S 2015 Wireless sensor networks for agriculture: The state-of-the-art in practice and future challenges *Comput. Electron. Agric.* 118 66–84

[9] Lerdsuwan P and Phunchongharn P 2017 An Energy-Efficient Transmission Framework for IoT Monitoring Systems in Precision Agriculture *Information Science and Applications 2017* Lecture Notes in Electrical Engineering vol 424, ed K Kim and N Joukov (Singapore: Springer Singapore) pp 714–21

[10] Gutierrez J, Villa-Medina J F, Nieto-Garibay A and Porta-Gandara M A 2014 Automated Irrigation System Using a Wireless Sensor Network and GPRS Module *IEEE Trans. Instrum. Meas.* 63 166–76

[11] Cambra C, Díaz J R and Lloret J 2015 Deployment and Performance Study of an Ad Hoc Network Protocol for Intelligent Video Sensing in Precision Agriculture *Ad-hoc Networks and Wireless* Lecture Notes in Computer Science vol 8629, ed M Garcia Pineda, J Lloret, S Papavassiliou, S Ruehrup and C B Westphall (Berlin, Heidelberg: Springer Berlin Heidelberg) pp 165–75

[12] Moghaddam M, Entekhabi D, Goykhman Y, Li K, Liu M, Mahajan A, Nayyar A, Shuman D and Teneketzis D 2010 A Wireless Soil Moisture Smart Sensor Web Using Physics-Based Optimal Control: Concept and Initial Demonstrations *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 3 522–35

[13] Baranwal T and Pateriya P K 2016 Development of IoT based Smart Security and Monitoring Devices for Agriculture 6
[14] Dan L, Xin C, Chongwei H and Liangliang J 2015 Intelligent Agriculture Greenhouse Environment Monitoring System Based on IOT Technology *2015 International Conference on Intelligent Transportation, Big Data and Smart City* 2015 International Conference on Intelligent Transportation, Big Data and Smart City pp 487–90
[15] Anon Homepage - LoRa Alliance®
[16] Nunamaker J F and Chen M Systems Development in Information Systems Research 10
[17] Putra S D, Sutikno S, Rosmansyah Y and Asrowardi I 2014 Integrated implementation of service and information security management system *2014 International Conference on ICT For Smart Society (ICISS)* 2014 International Conference on ICT For Smart Society (ICISS) pp 185–91
[18] Sereati C O, Sumari A D W, Adiono T and Ahmad A S 2017 Cognitive artificial intelligence (CAI) software based on knowledge growing system (KGS) for diagnosing heart block and arrythmia *2017 6th International Conference on Electrical Engineering and Informatics (ICEEI)* 2017 6th International Conference on Electrical Engineering and Informatics (ICEEI) pp 1–5.